

Module 11: Join Dependencies (5NF) & Inclusion Rules

Join Dependencies (JD), 5NF & Inclusion Dependencies

Project-Join Normal Form (PJNF), N-Way Lossless Join & Master Normal Form Hierarchy

1. Join Dependency (JD) & 5NF (PJNF)

N-way lossless decomposition beyond binary pairs

Join Dependency Definition: $\bowtie[R_1, R_2, \dots, R_n]$
Relation R satisfies JD $\bowtie[R_1, R_2, \dots, R_n]$ iff:
 $R = \pi_{R_1}(R) \bowtie \pi_{R_2}(R) \bowtie \dots \bowtie \pi_{R_n}(R)$

Fifth Normal Form (5NF / PJNF) Criteria:
Relation R is in 5NF iff it is in 4NF and for every non-trivial Join Dependency $\bowtie[R_1, R_2, \dots, R_n]$:
EVERY R_i MUST BE A SUPER KEY OF R.
If this holds, table cannot be losslessly decomposed any further!

Inclusion Dependencies (IND): $R[X] \subseteq S[Y]$
Subset constraints between relations.
Every value appearing in column X of R must also appear in column Y of S.
★ Direct formal basis of FOREIGN KEY and Referential Integrity!

2. Master Normal Form Hierarchy

Progressive Constraint Nesting: $5NF \subset 4NF \subset BCNF \subset 3NF \subset 2NF \subset 1NF$

1NF: Atomic Domains
Eliminates multi-valued attributes and composite attributes.

2NF: No Partial Dependencies
Non-prime attributes must depend on the FULL candidate key.

3NF: No Transitive Dependencies
For $X \rightarrow Y$: X is Super Key OR Y is Prime. Lossless + Preserved!

BCNF: Strict Super Key Determinant
For all $X \rightarrow Y$: X MUST be Super Key. May lose dependency preservation.

4NF: No Multi-Valued Dependencies
For all $X \twoheadrightarrow Y$: X is Super Key. Eliminates cartesian explosion.

5NF: No Join Dependencies (PJNF)
All non-trivial JDs implied by candidate keys. Zero join redundancy.

> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Join Dependencies $\bowtie[R_1, \dots, R_n]$, Fifth Normal Form (5NF / Project-Join Normal Form PJNF), Cyclic Constraint Dependencies, Inclusion Dependencies (IND) and Foreign Key Relationship, Master Normal Form Summary.

1. Limitations of 4NF: The 3-Way Cyclic Join

4NF sirf do independent multi-valued attributes ke binary splits ko handle karta hai (R

$\rightarrow R_1, R_2$).

Lekin database design me aisi cyclic relationships ho sakti hain jinhe do tables me split karne par **lossless join nahi milta**, lekin **teen ya usse zyada tables me split karne par lossless join mil jaata hai!**

Is generalisation ko capture karne ke liye **Join Dependency (JD)** define ki jaati hai.

2. Join Dependency (JD) Formal Definition

Ek relation schema R par **Join Dependency** $\bowtie[R_1, R_2, \dots, R_n]$ tab hold karti hai jab relation R apne projections ke natural join ke strictly equal ho:

$$R = \pi_{R_1}(R) \bowtie \pi_{R_2}(R) \bowtie \dots \bowtie \pi_{R_n}(R)$$

Where each $R_i \subseteq R$ and $\bigcup_{i=1}^n R_i = R$.

Trivial Join Dependency: Agar kisi ek $R_i = R$ ho jaye, toh JD trivial hoti hai.

3. Fifth Normal Form (5NF / Project-Join Normal Form - PJNF)

3.1 Formal Definition

Ek relation schema R **Fifth Normal Form (5NF)** me tab hota hai jab:

1. Wo relation already **4NF** me ho.
2. Har non-trivial Join Dependency $\bowtie[R_1, R_2, \dots, R_n]$ ke liye:

$\mathbf{\{ \text{EVERY } R_i \text{ IS A SUPER KEY OF } R \}}$

3.2 Significance

5NF relational database theory ki ultimate normal form maani jaati hai (with respect to projection and join operators).

Agar koi table 5NF me hai, toh use aage kisi bhi number of tables me bina spurious tuples ke split nahi kiya ja sakta jab tak un tables ka key original relation ka super key na ho!

4. Inclusion Dependencies (IND)

4.1 Formal Definition

Given two relation schemas R and S , attribute sequences $X \subseteq R$ and $Y \subseteq S$ (with matching domain and length):

$\mathbf{R[X] \subseteq S[Y]}$

Iska matlab hai ki relation R ke kisi bhi tuple me attribute X ki jo values aayengi, unka relation S ke attribute Y me pehle se present hona mandatory hai!

4.2 Connection to Foreign Key Constraints

Inclusion dependency relational database me **Foreign Key (Referential Integrity)** ka formal theoretical foundation hai!

SQL syntax:

```
`sql
```

```
FOREIGN KEY (Dept_ID) REFERENCES DEPARTMENT(Dept_ID);
```

,

Is directly the Inclusion Dependency: $\text{EMPLOYEE}[\text{Dept_ID}] \subseteq \text{DEPARTMENT}[\text{Dept_ID}]$.

5. Architectural Diagram

6. Master Summary of All Normal Forms (AKTU Quick Reference Table)

Normal Form	Core Requirement	Violation Removed	Lossless Join?	Dependency Preserving?
1NF	Atomic attribute values only	Multi-valued & composite attributes	\$\checkmark\$	\$\checkmark\$
2NF	In 1NF + No Partial Dependencies	Non-prime depending on proper subset of key	\$\checkmark\$	\$\checkmark\$
3NF				

ightarrow Y\$: \$X\$ is Super Key **OR** \$Y\$ is Prime | Transitive dependencies (Non-prime \$

ightarrow\$ Non-prime) | \$\checkmark\$ | **\$\checkmark\$ (Always Guaranteed!) |**

BCNF

ightarrow Y\$: \$X\$ **MUST** be Super Key | Overlapping candidate keys redundancy | \$\checkmark\$ | \$imes\$ (May be lost) |

4NF	In BCNF + For every \$X\$ woheadrightarrow Y\$: \$X\$ is Super Key	Independent multi-valued attributes (Cartesian product)	\$\checkmark\$	N/A (MVD focus)
5NF (PJNF)	In 4NF + Every \$R_i\$ in JD \$owtie[R_1..R_n]\$ is Super Key	N-way join dependencies & spurious tuples	\$\checkmark\$	N/A